1 Cover

- Salam, naam, project title
- Supervisor: Dr. Aneela Zameer
- Startup ke problem se inspired, academic prototype, sirf open data

2 Background

- Cane: **3M+ acres**, sugar **2nd** bari agro industry
- Agritech startup link: **33 mills** waiting list, kuch ko prototype diya
- JDW: major client aur investor, **4.5 lakh acres** digitised, agronomists use karte hain
- Open data pe academic prototype

3 Problem

- Mills: monitoring = **manual field surveys**, satellite-based per-field nahi
- Milling cycle se pehle: yield ke hisab se planning aur price
- Humara project: leadership ko field monitoring se faislay mein madad
- 13-country map: Pakistan mein **recall 43%** (local labels nahi)
- SUPARCO: satellite-based govt monitoring ka **main idara**, statistics ke liye

4 Aim

- Web portal: digitised fields, crop labels, health history
- Vigour (**NDVI**) vs normal cane cycle → health
- Official yields + indices → **regression** → indicative yield
- Disease diagnose **nahi**, sirf stress ke effects
- Per-field yield **dikhayenge**, validate nahi: field-level ground truth nahi
- NFRs: non-professionals ke liye aasan; **modest hardware** (laptop, Colab, Kaggle); map **~5,600 fields** pe smooth

5 Interface

- Agronomist ke saare monitoring tools aik jagah
- Date slider → purani images ka analysis
- **Mockup hai, asal results nahi**

6 Existing systems

- SUPARCO: government statistics; yield models **1 km NDVI**; mills ke liye tool nahi
- Cropin, OneSoil: closed source, Pakistan ke liye tailored nahi

7 Research

- 13-country map: Pakistan mein **recall 43%**, local labels nahi
- Punjab study: deep learning achha, lekin sirf **~130 acres**; dataset request ki
- Satellite + weather → yield; France: time-series deep models > Random Forest
- In findings se 3 modules → aik decision-support system

8 Method

- Pakistan mein zaroori open data nahi
- SUPARCO aur Punjab study ke authors se contact kiya
- Company: **8 lakh acres** digitised (JDW ke 4.5 lakh samait); policy ki wajah se FYP mein use nahi
- Iterative, Agile: plan, approve, build, review

9 Datasets

- CSB boundaries; CDL labels: **95% producer's**, **96% user's accuracy**
- Sentinel-2 images, NASS official yields
- **18 seasons** (2008–2025): **temporal hold-out**
- La Réunion, Brazil, EuroCrops dekhe: kuch missing tha

10 Architecture

- Sources: imagery, boundaries + labels, yield + weather
- Ingestion: common interface (**adapter pattern**), source badla ja sakta hai
- Common **CRS**, **field = single unit**, poori images store nahi
- 3 models: crop, health, yield → web map (layers, slider, pop-up)

11 Use cases

- Mill officer / agronomist: map, field click (crop, health, yield), date change
- Developer: pipeline, models evaluate, naye regions ingest

12 Data model

- Sab kuch **FIELD** se juda: labels, observations, predictions (saal-ba-saal)
- REGION → FIELD: **one-to-many**; region ke official yield statistics

13 Tech stack

- Open aur free: Python geo libraries, Planetary Computer
- GeoParquet: column-based; humari files mein row = field
- DuckDB: **embedded (in-process)** analytical DB, server nahi
- **Random Forest** baseline → TempCNN, attention (PyTorch); MapLibre; Earth Engine optional

14 Progress

- Data checks pehle, **acceptance criteria** pehle se set
- Done: environment, literature, datasets
- Ab: preprocessing

15 Timeline

- Sem 7 done: environment, data selection, fields aur labels checked
- Sem 7 baqi: imagery aur yield checks, literature round 2, model selection
- Architecture final; models experiments se
- Sem 8: Modules A, B, C, web map → Pakistan stage, thesis, defence
- Courses throughout: ML aur geospatial SWE

16 Thanks

- Shukriya, sawalat
- Tayyari: US data kyun? Labels sahi kaise? SUPARCO aur Cropin se farq? Health ka matlab? Server kyun nahi?